# Questions

**4.47** A zener diode is rated for 3-V operation at 5 mA with an incremental resistance of 100 Ω. It is connected to a voltage source $V_S$ through a 1-kΩ resistor. What is the maximum voltage $V_S$ that can be applied without exceeding the zener's maximum power rating of 40 mW? What is the voltage across the zener under this condition?

**4.86** A clamped capacitor using an ideal diode with cathode grounded is supplied with a sine wave of 5-V rms. What is the average (dc) value of the resulting output?

**4.87** The clamped capacitor in Fig. P4.87 is loaded by the resistor $R$. The input $v_I$ is a symmetric 10-V peak-to-peak square wave at 100 kHz. Sketch the waveform at $v_O$ if the time constant $CR = 10\,\mu s$.

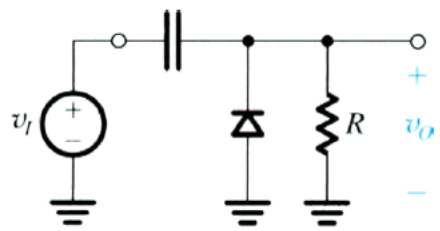



Figure P4.87

***4.88** For the circuits in Fig. P4.88, each utilizing an ideal diode (or diodes), sketch the output for the input shown. Label the most positive and most negative output levels. Assume $CR \gg T$.

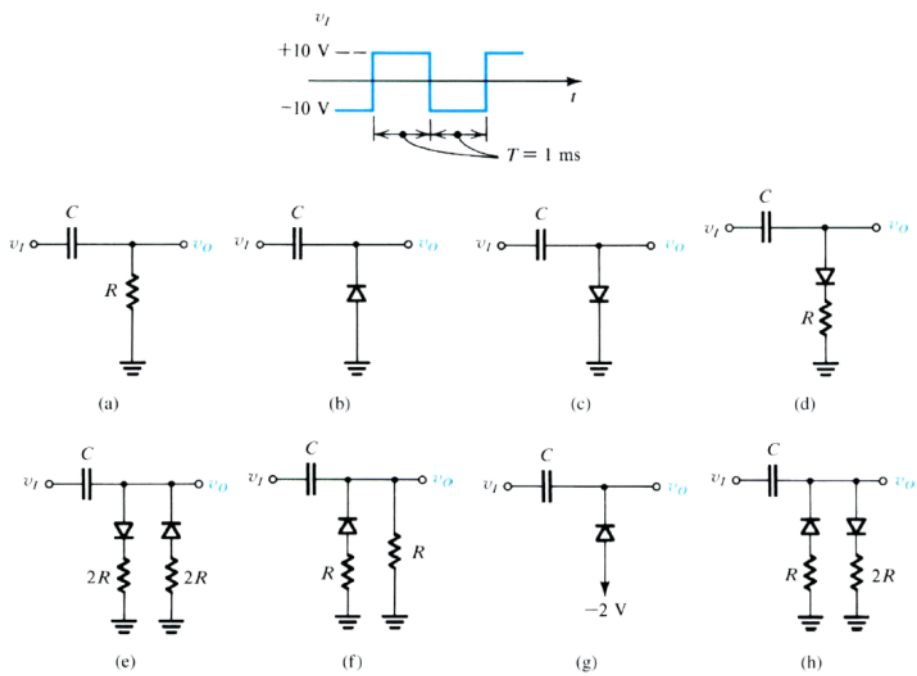



Figure P4.88